

Poke / force / release

Sometimes you need to deposit a value while the sim is live

Soft poke vs sticky force

- Prefer poke when you want a temporary live value
- Prefer force only when you must hold a net against its drivers
- Reset nets can be held for debug; data nets are the usual poke or force targets
- Clocks are a special hazard: forcing a clock is unsafe and easy to forget

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Poke / force / release

Document-aid literacy: **poke** (soft live drive) vs **force** / **release** hazards (not a full simulator). Starter: poke + force→release on data — READY.

Starter example: poke + force→release on data — no forces left — READY.

Load starter example

Challenges 0 / 22 cleared

Quiz: poke: Poke means...

☐ a soft live deposit for the next eval (does not stick like force)

☐ a permanent override until reboot

☐ only VCD export

☐ enabling -Wall

Show hint

Check

Next

Idle

1

2

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Core ideas

Poke

Soft live deposit for the next eval.

Force

Hard override until release.

Release

Clear force; drivers resume.

Hazard

Stuck force · forcing clocks.

Lab

SCENARIO

DEPOSIT VALUE

ready starter

▼

1

▼

Load preset

Poke

Force

Release

Scan

Lab starter

learn_hdl_simulator — hdl sim poke force

Public simulator practice

- In the public IDE, poke a data net once and observe the wave
- Force the same net, run a little, then release and confirm drivers resume
- Avoid forcing the clock
- If the tool warns about active forces, treat that as a first-class debug clue, not noise

Pitfalls to watch

- Do not leave forces active and then “debug” wrong RTL
- Do not force clocks to “make time move.” Do not confuse poke with force
- And always know which signal you selected before you deposit a value

Your turn

- Complete the checklist for at least one track, preferably both
- Practice poke, force, and release, and leave the session with no active forces
- When you are ready, take the short quiz, then continue to full-sim waves